

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

เลขที่นั่งสอบ

Midterm Examination of Semester 2

Year: 2012

Course: 340152 Electrical Measuring Instruments

Section: 5-6

Date: 5 January 2013

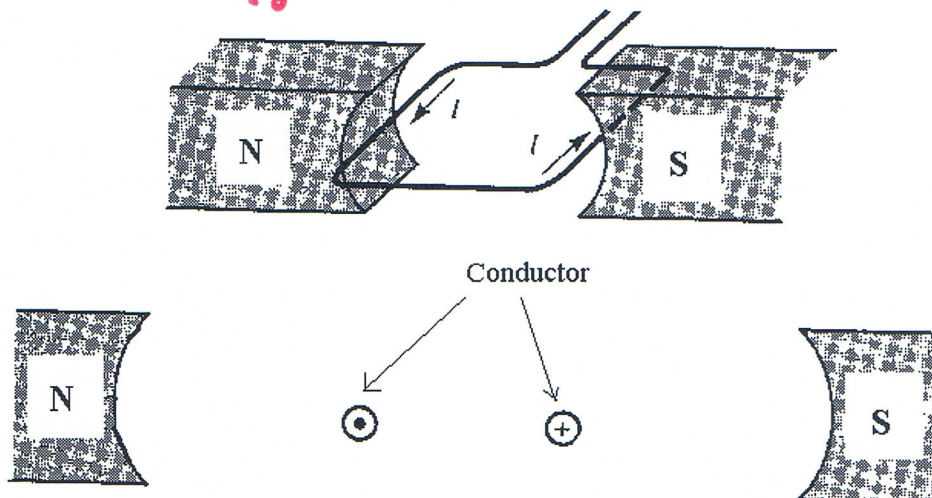
Time: 10.00-12.00

Name _____ ID. _____ Class _____

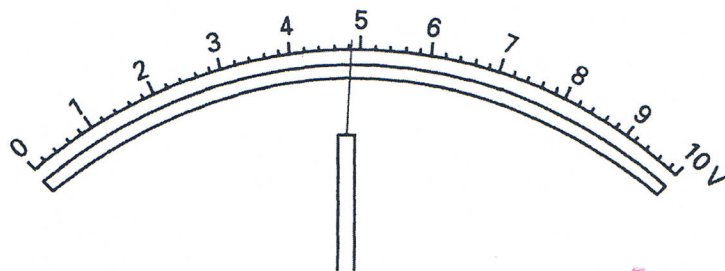
Instructions:

1. Cheating will result in all classes registered for the current semester.
Student who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. This will be a close book examination.
4. Calculation devices are allowed.
5. Dictionaries are allowed.
6. Electronic communication devices are NOT allowed in the examination room.
7. The exam paper consists of 7 pages with 12 questions and a total score of 70 marks.
8. Write all your answers in this examination sheet.

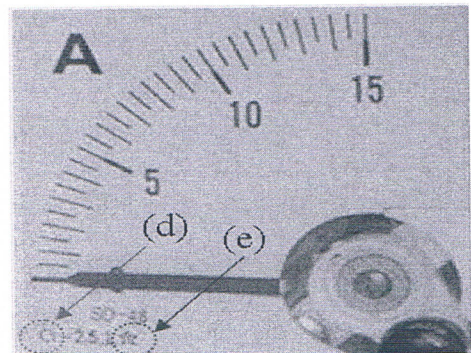
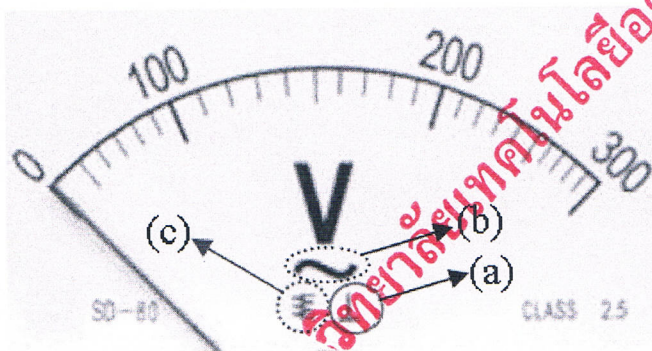
Q1) Describe the briefly the principle of the PMMC movement by using **Right Hand Grip Rule** to sketch the direction of the magnetic field and force. (4 marks)



Q2) The analog voltmeter in Figure have a typical accuracy of $\pm 2\%$ of full scale, or $\pm 2\%$ of 10 V. Thus, the measured quantity is V, meaning that the actual voltage is somewhere between V and V. So, the measurement precision of analog instrument is.....V (4 marks)



Q3) Identify each of the symbols illustrated in Figures. (5 marks)



a)

b)

c)

d)

e)

Q4) Define the following terms:

(5 marks)

(a) Measurement :

(b) Instrument :

(c) Error :

(d) Accuracy :

(e) Precision :

Q5) List three basic functions of instruments

(3 marks)

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Q6) List three advantages that electronic instruments have over electronic instruments.

(3 marks)

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Q7) Explain Gross, Systematic, and Random Errors.

(3 marks)

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Q8) Explain the function of each force involved in the moving system of a deflection

instrument.

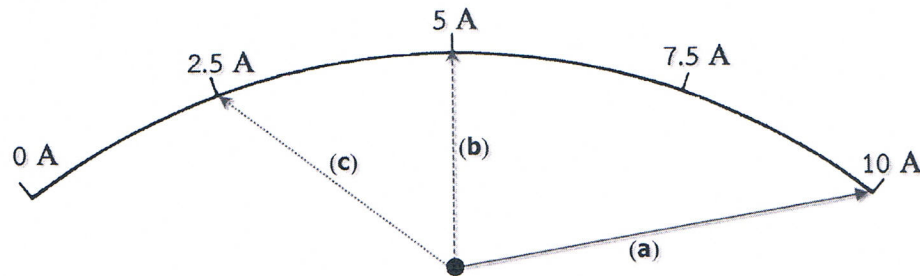
(3 marks)

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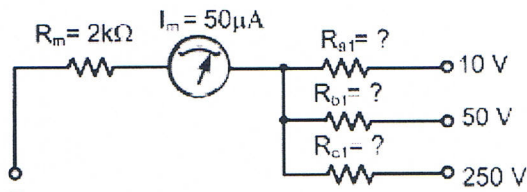
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Q9) An Ammeter range 10A class 1.5, Calculate the maximum %Error for 3 different position of pointer Case (a) = 10A, Case (b) = 5A, Case (c) = 2.5A. (10 marks)

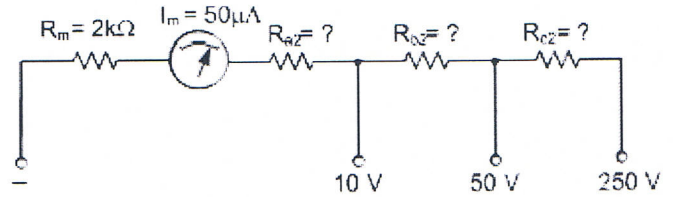


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Q10) A PMMC instrument with FSD = $50\ \mu\text{A}$ and $R_m = 2\ \text{k}\Omega$, is to be employed as a **volt-meter** with ranges of 10 V, 50 V and 250 V. **Calculate** the required values of multiplier resistors(R_a , R_b , R_c) for both cases and give the **comparison** to the practical volt-meter. (10 marks)



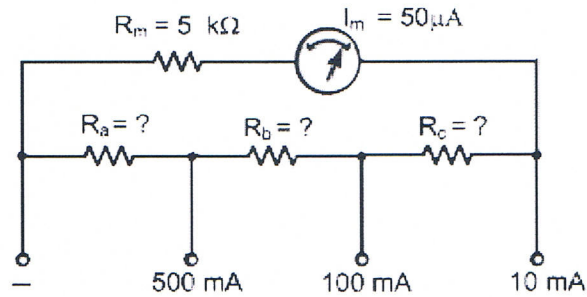
Case (a)



Case (b)

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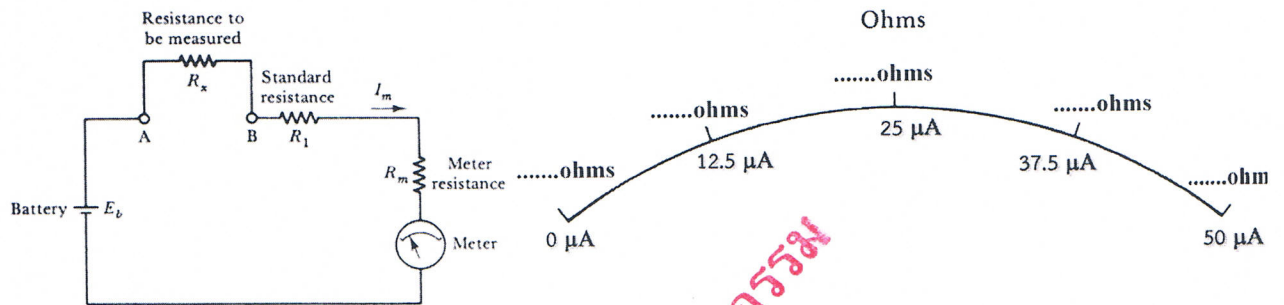
Q11) A PMMC instrument has $50\ \mu\text{A}$ and a coil resistance of $5\ \text{k}\Omega$. Calculate the required resistance (R_a , R_b , R_c) values to convert the instrument into **an ammeter** (multi-range) with an FSD of 500 mA, 100 mA, and 10 mA. (10 marks)



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Q12) The series ohmmeter in figure is made up of a 1.5 V battery, a 50 μA meter, $R_m = 2 \text{ k}\Omega$.

- (a) Calculate the value of R_1 to make the current to be full scale.
- (b) Determine how the resistance scale should be marked at $R_x = 0$ and $R_x = \infty$.
- (c) Determine how the resistance scale should be marked at 0.75 FSD, 0.5 FSD, and 0.25 FSD. (10 marks)



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